Figure. Three-dimensional dimensions of the suture pad. The wound is positioned at the center of the pad and has an opening at the surface **(a).** A cross-sectional front view of the pad is shown in **(b).** The wound was modeled as a wedge-shaped defect with its maximum opening at the surface and a constant depth of 0.5 cm. The hole depth and diameter were fixed at 0.4 cm and 0.7 mm, respectively. The forces generated during knot tying were simulated as pressures applied to the internal surfaces of the holes, as illustrated in the front view **(c).** These pressures were applied along the line connecting the centers of each pair of holes (corresponding to the needle entry and exit points of a single suture), as shown in the top view **(d).**